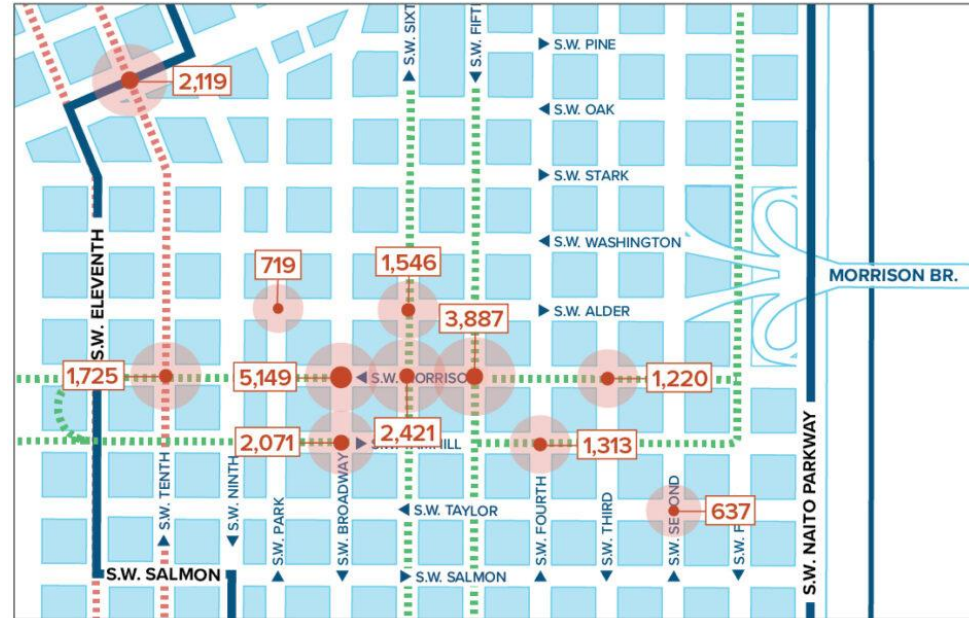


Improving Pedestrian Count Estimation With Machine Learning and Data Fusion

Naomi Cai

Introduction

- Pedestrian volume data and estimation helps inform urban planners
- Important for development of safety measures, infrastructure, etc.



Source: Downtown Portland Clean & Safe

Background pt. 1

- Data is typically measured at intersections with automatic or manual counters

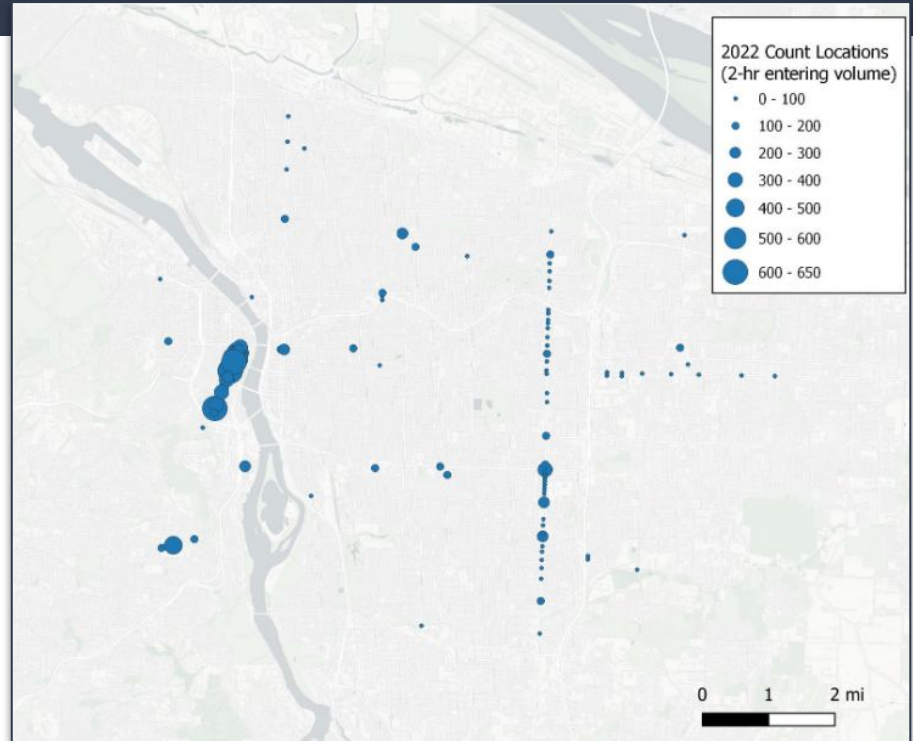


Intersection was defined as 30m area around count location

Source: Kothuri et al.

Background pt. 2

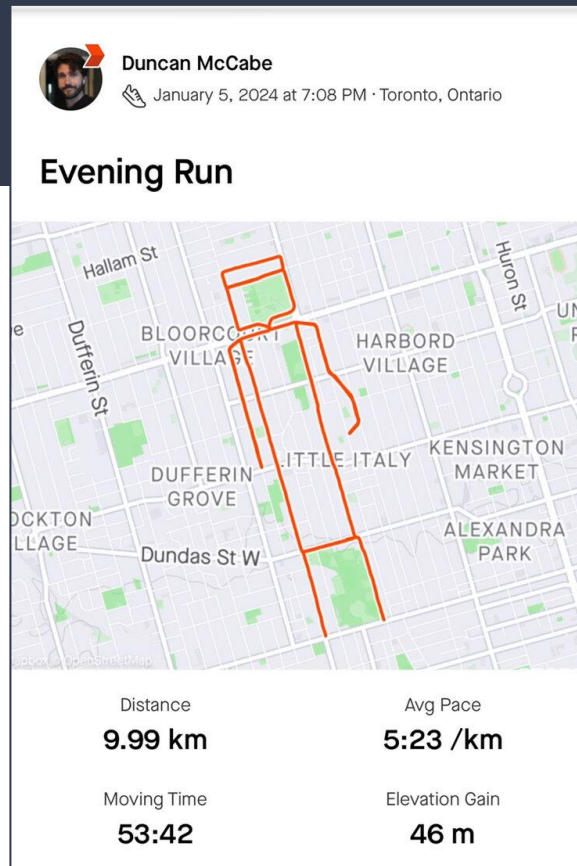
- Both forms of counters have downsides, namely cost and scalability
- Need for more convenient ways to model this



Source: Kothuri et al.

Background pt. 3

- Strava is a fitness app, often used as an activity tracker
- Previous work was done by Kothuri et al., using this data + a negative binomial regression to model pedestrian count



Source: New York Times

Goal

Considering the limitations of current methods, our goal is to apply machine learning techniques to hopefully improve pedestrian count models and achieve better performance over simpler statistical models

- initial plan: test and compare 4 methods, using the previous negative binomial regression model as a baseline for performance

Data

- 3 components: counter, strava, and static data



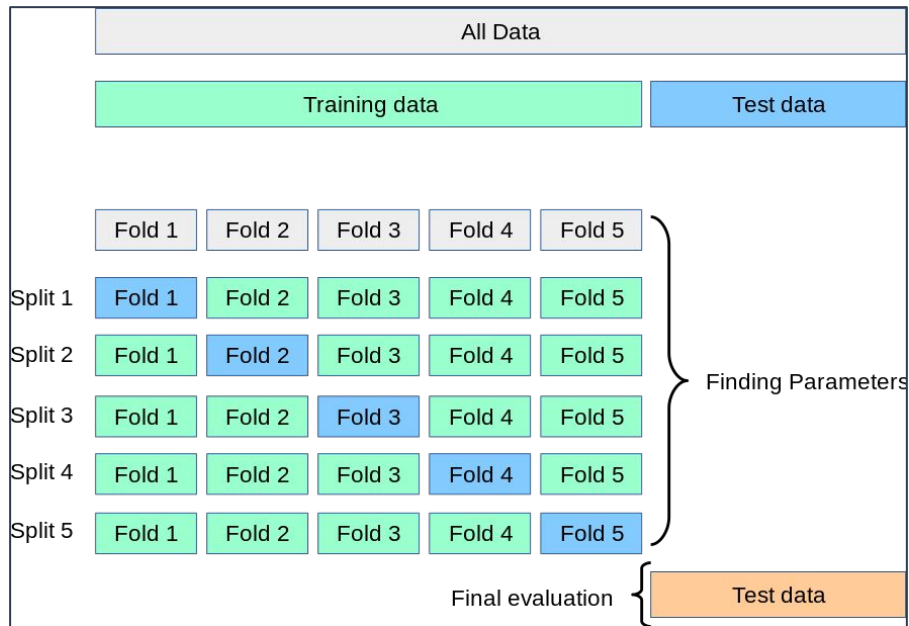
Source: Kothuri et al.

Methods pt. 1

- Initially tried XGBoost, Feedforward neural network (FNN), ElasticNet regression, Support Vector Regression (SVR)
- Code was written in Python using typical ML/data science packages (Pandas, sklearn, Pytorch, etc.)

Methods pt. 2

- Data divided into 80-20 train test split
- 5 fold cross validation used for hyperparameter tuning, then models were trained with optimal hyperparameters on the total train set, and RMSE, MAE calculated

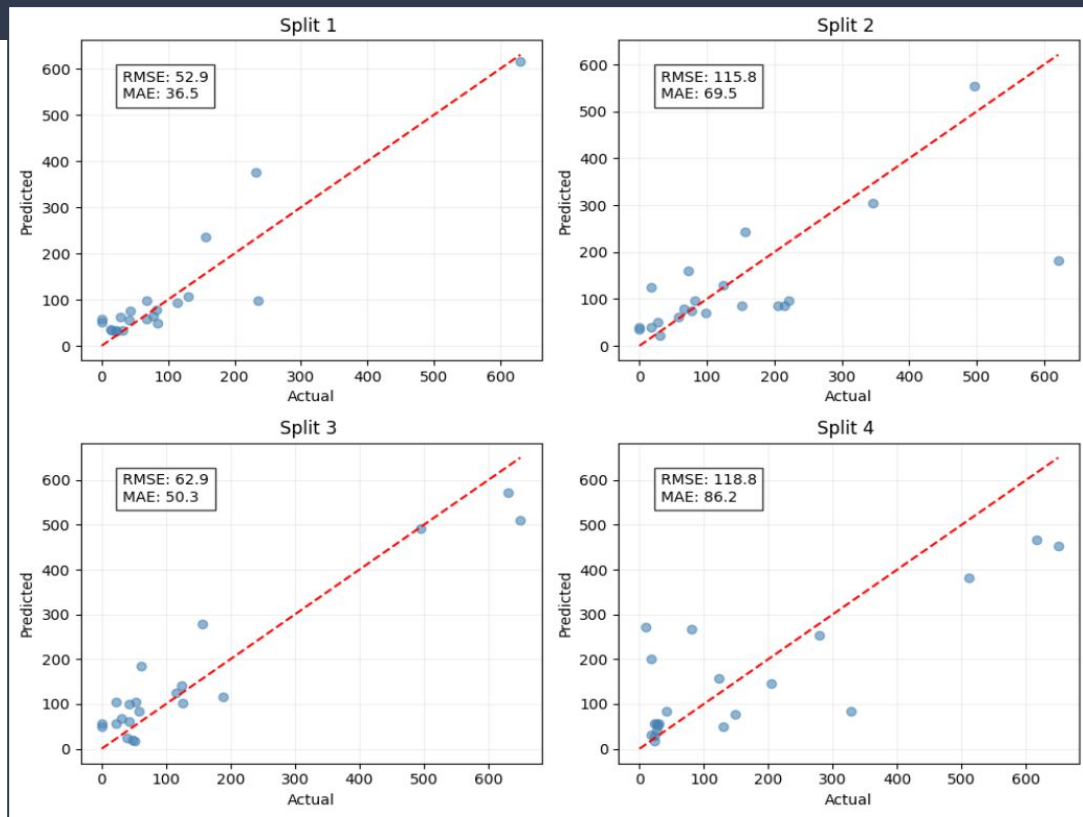


Source: Analytics Yogi

Preliminary Results pt. 1

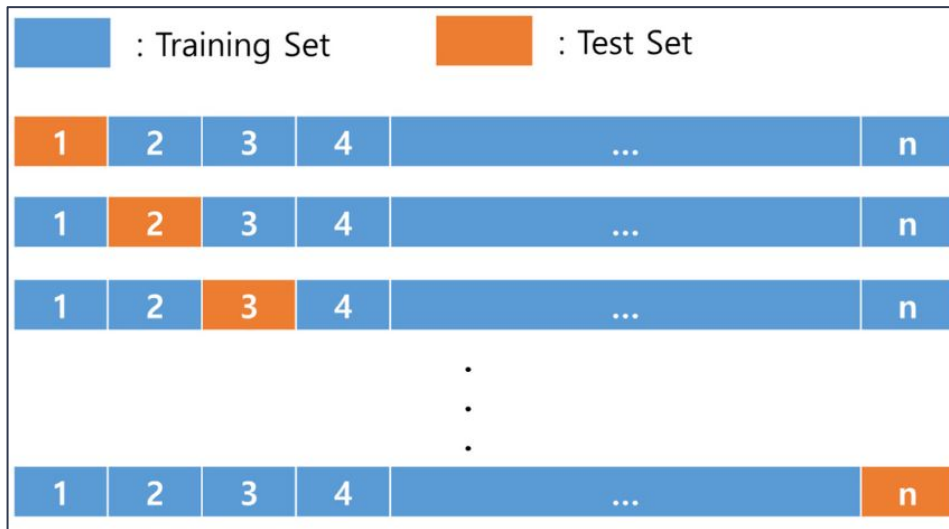
	RMSE	MAE
Negative Binomial	103.83	65.37
ElasticNet	64.47	38.89
XGBoost	48.24	32.46
ANN	99.26	61.88
SVR	81.77	47.28

Preliminary Results pt. 2



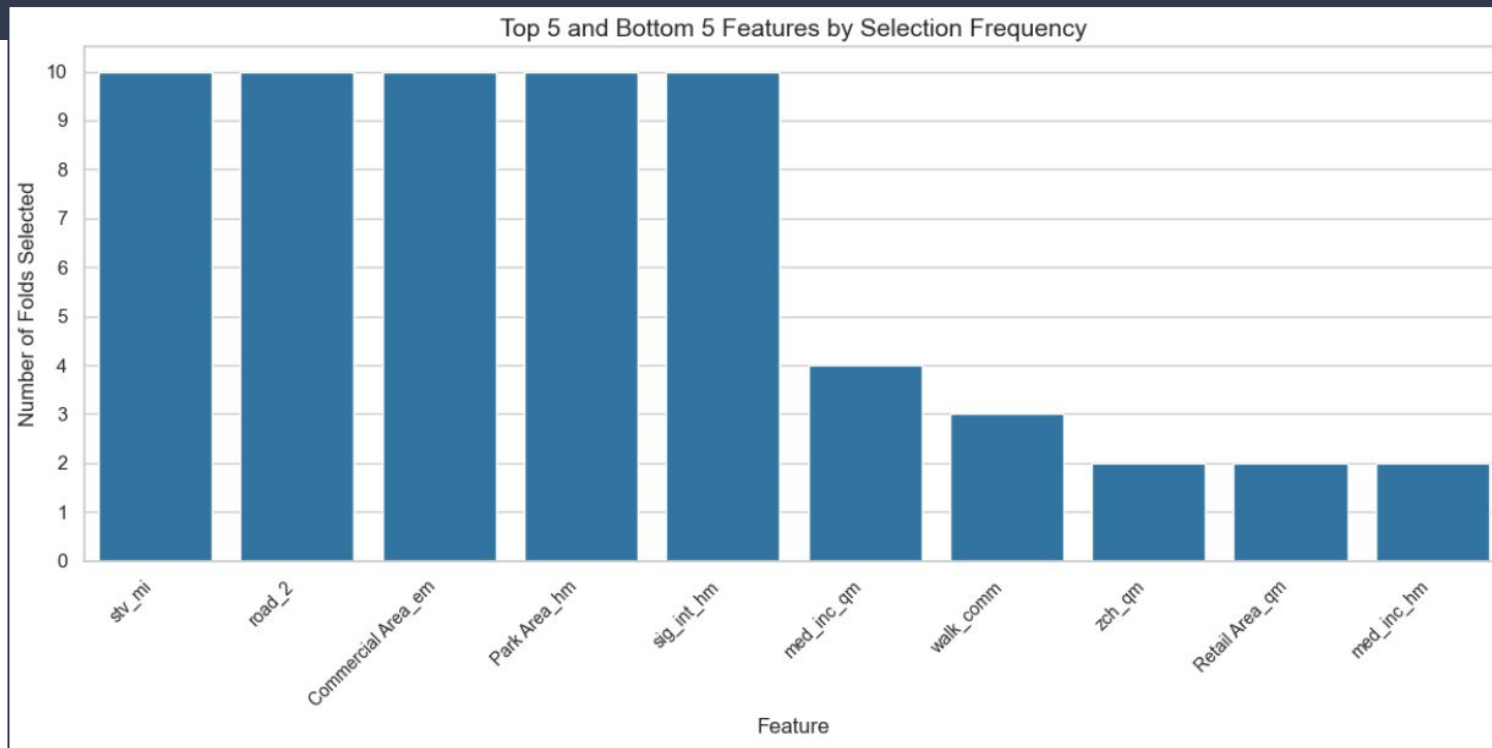
Methods pt. 3

- Unstable measures of performance: use leave one out cross validation (LOOCV)
- Hyperparameter tuning yielded little to no improvement

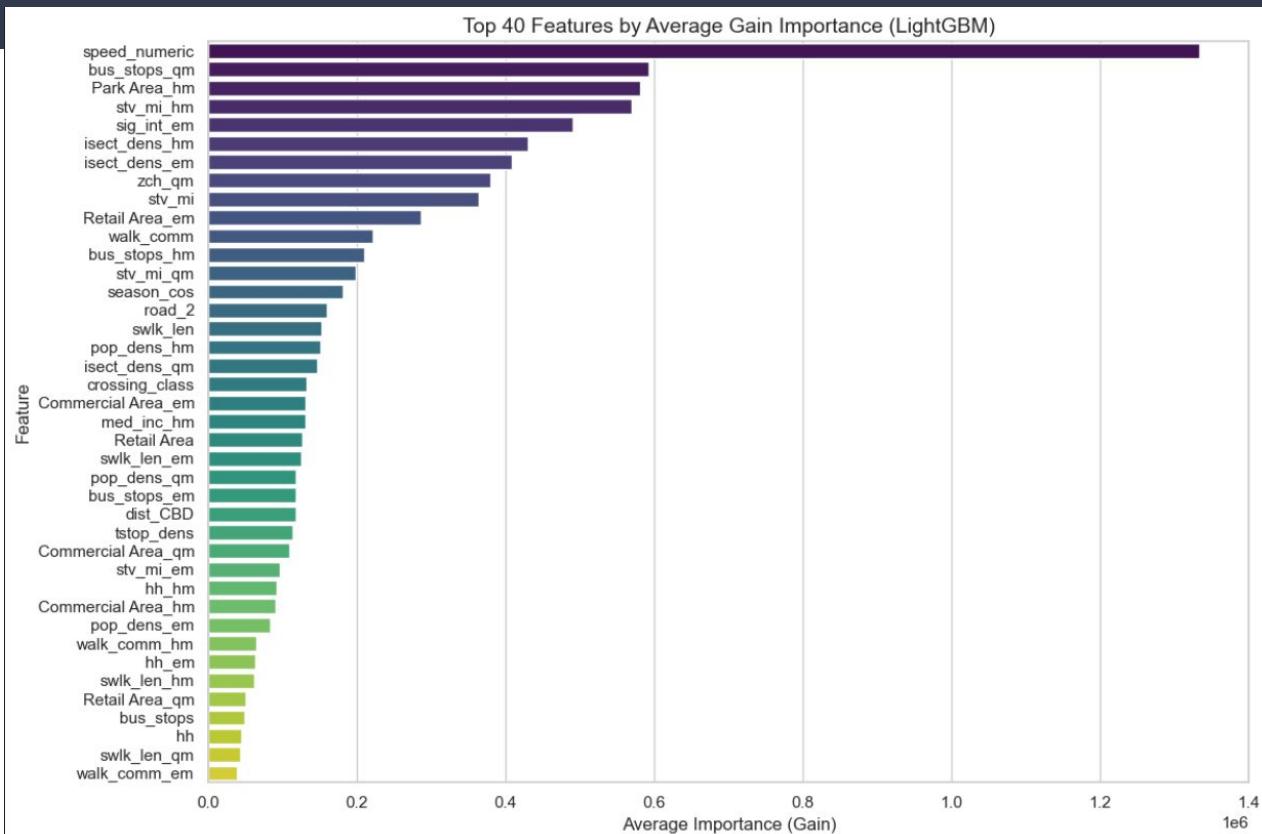


Source: Cha et al.

Methods pt. 4



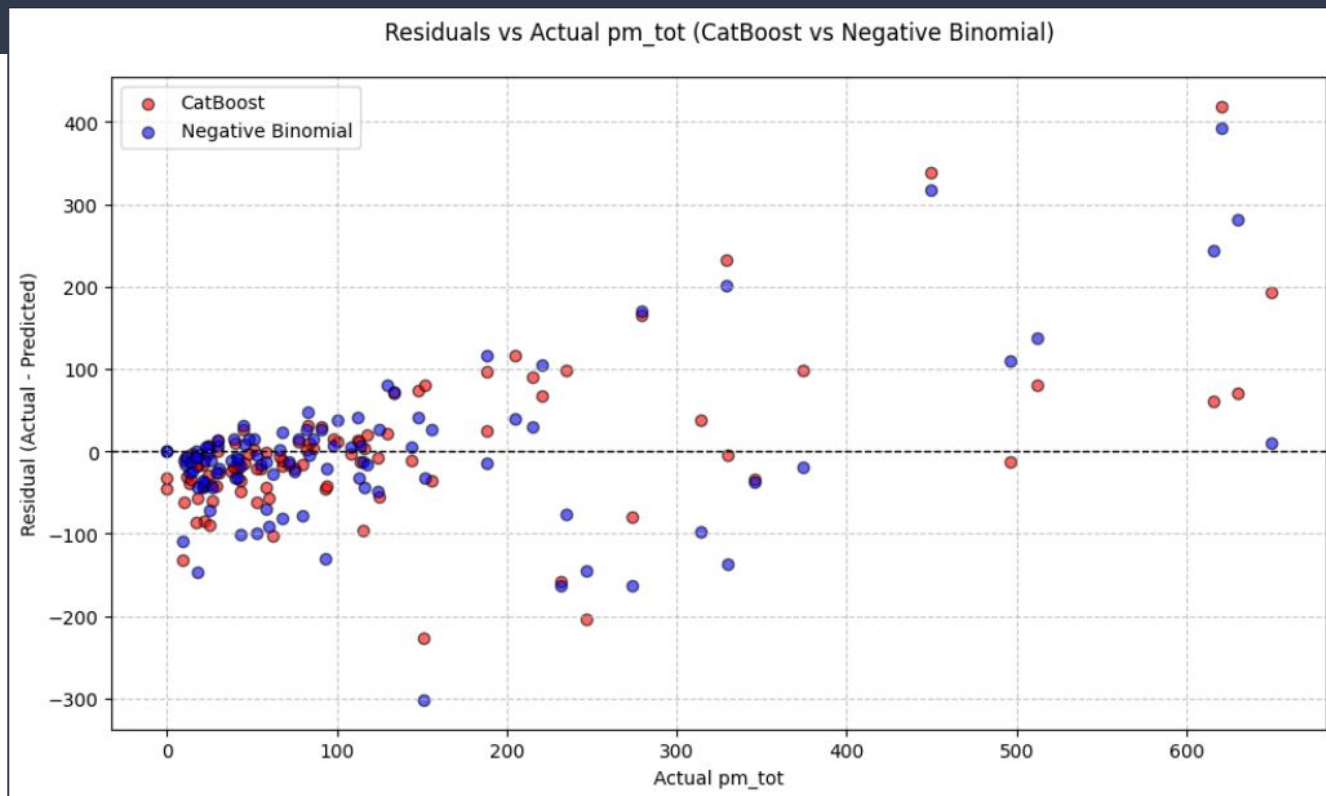
Methods pt. 5



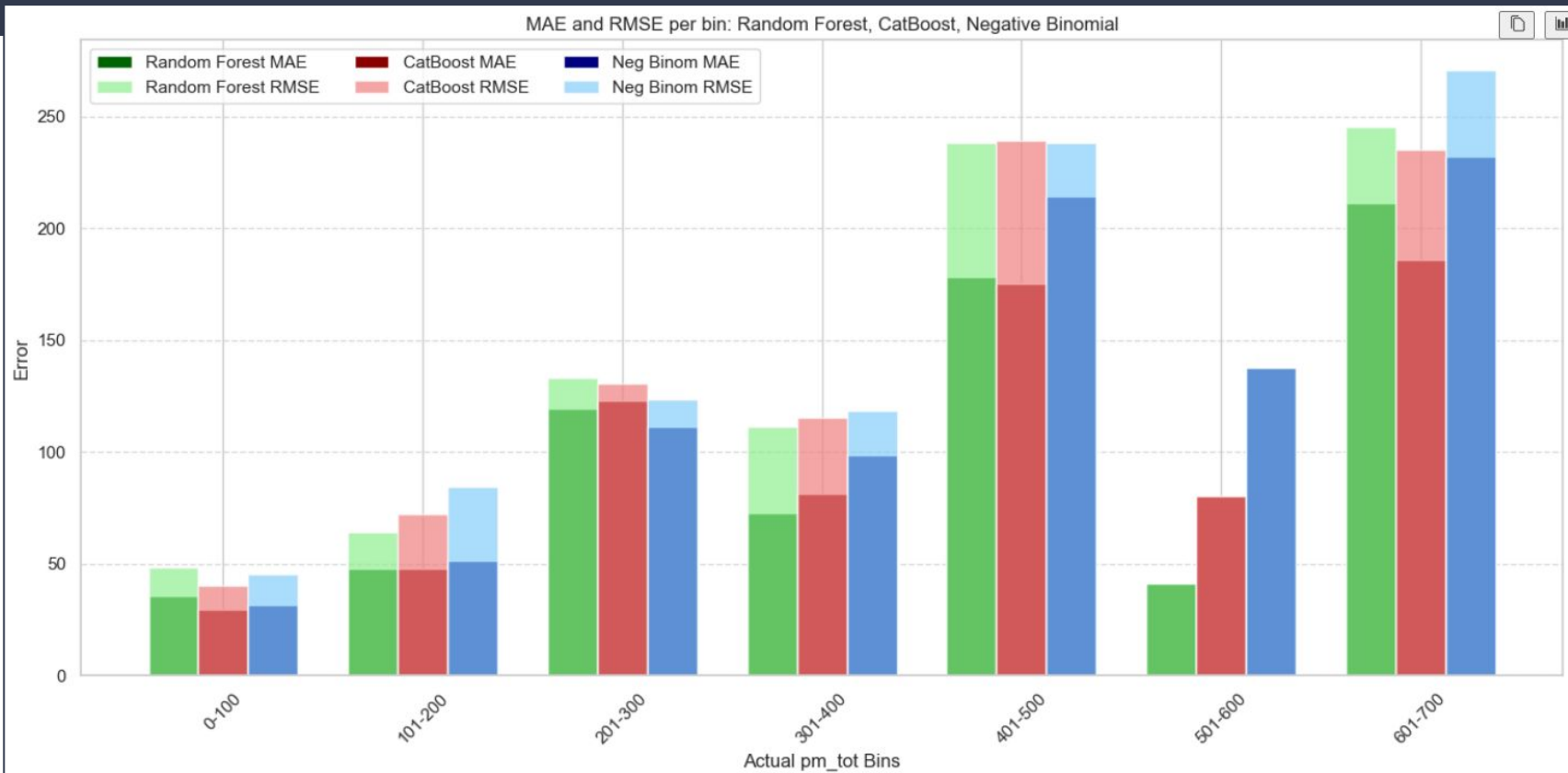
Results

	RMSE	MAE
Negative Binomial	92.91	56.59
CatBoost	85.58	52.67
RandomForest	87.58	56.50

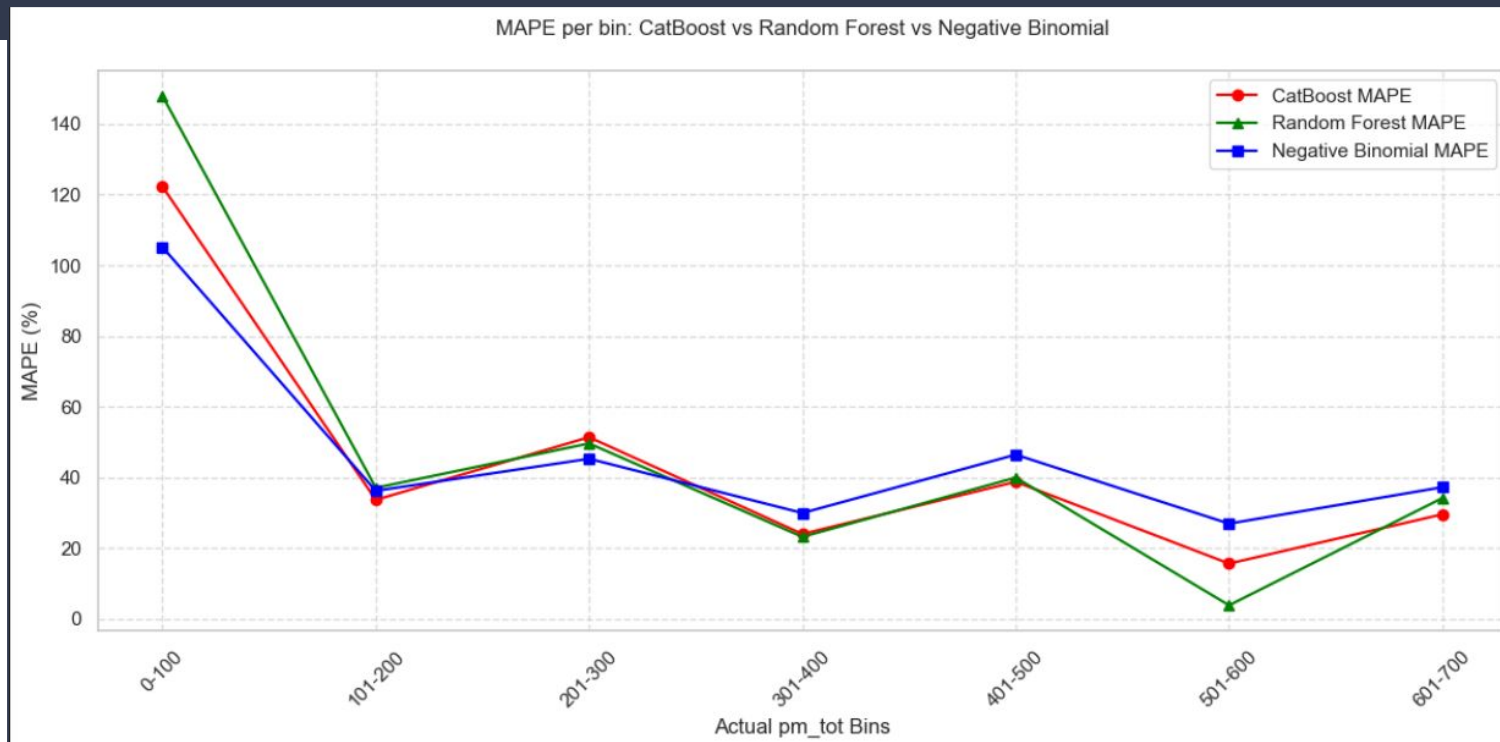
Analysis pt. 1



Analysis pt. 2

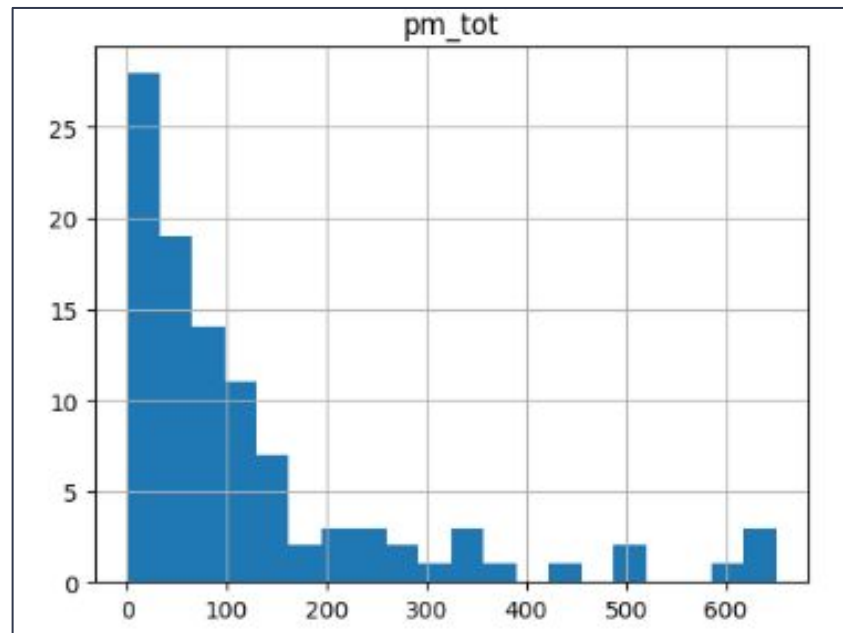


Analysis pt. 3



Conclusion

- Very small dataset + highly skewed
- In general, it is more important for a model to perform well in high count areas



Takeaways

- Both random forest and catboost methods tended to slightly outperform the baseline regression
- Suggests that ML techniques have the potential to boost pedestrian volume modelling, especially if more data is available

Acknowledgements

Sirisha Kothuri

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